MIL-P-28636B
15 February 1985
SUPERSEDING
MIL-P-28636A
23 January 1979

MILITARY SPECIFICATION

PANEL FOR BUNK BED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for laminated plastic surfaced particle board bunk bed headboard and footboard panels.

1.2 Classification. The bunk bed panels shall be as shown on figures 1 and 2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-508 - Plastic Sheet, Laminated, Decorative and Nondecorative.
UU-P-268 - Paper, Kraft, Wrapping.
MMM-A-188 - Adhesive, Urea-Resin-type (Liquid and Powder).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-15228 - Bed, Bunk, Steel, Single and Double Deck (Convertible).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

U.S. DEPARTMENT OF COMMERCE

Commercial Standard CS 236 - Mat-Formed Wood Particle board.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bunk bed panel shall consist essentially of a particle board core surfaced on each side with a decorative laminated plastic sheet, a tee-style plastic edge molding, and four clamps, capped teenuts, and machine screws. Two panels, one to be used as a headboard and one footboard, are required for each bunk bed.

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3.2 First article. Unless otherwise specified (see 6.2), the contractor shall furnish a panel for first article inspection and approval (see 4.2.1 and 6.3). The first article inspection shall be performed at the manufacturer's plant or at the installation site, as specified (see 6.2). The first article shall be a sample panel produced prior to initiation of the production items.

3.3 Interchangeability. All parts shall be manufactured to standards that will permit replacement or adjustment without modification of parts. All threaded parts shall conform to Handbook H-28.

3.4 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.4.1 Particle board. Particle board shall be 3/4-inch thick and in accordance with CS 236, type 1, grade B, class 2.

3.4.2 Decorative laminated plastic sheets. Decorative laminated plastic sheets shall be 0.032-inch thick and in accordance with L-P-508, style D, type I, class 1. Pattern, color, and finish shall be as specified (see 6.2).

3.4.3 Plastic edge molding. Plastic edge molding shall be tee-style, extruded of ethyle cellulose, polupropylene, or cellulose acetate butyrate. The tongue of the tee shall be formed with barbs to hold securely in the slot of panel edge. Face shall be convex, and smooth without grooves or other detailing. Edge molding shall either match the wood grain of the decorative laminated plastic sheet or be a solid color which blends with the woodgrain color, as specified (see 6.2). When solid color laminated plastic sheets are used, the edge molding color shall be of the same color or of the same color of a lighter shade.

3.4.4 Clamps. Clamps for securing the panels to the bed cross tubes, as shown on figure 1, shall be formed of low-carbon cold or hot-rolled, pickled, and oiled steel, C-1010, 5/8-inch wide, and not less than 0.048-inch thick. Clamp shall be a full-loop closed clip, but furnished open for slipping over the 7/8-inch diameter cross tubes. Opening in each end of the clamp shall be of proper diameter to accommodate the machine screw specified in 3.4.6. Clamp finish shall be bright zinc or cadmium plated and the loop shall be coated with a nominal 1/32-inch thick dispersion of vinyl chloride based resins in liquid plasticizers, polyvinyl chloride (PVC). Unless otherwise specified (see 6.2), the PVC coating color shall be FED-STD-595 No. 16251, gray. The clamp shall be similar and equal to Model No. COV-1509 of KMC Stampings, Divison of Kickhaefer Manufacturing Co. 1221 S. Park Street, Port Washington, WI 53074.

3.4.5 Label holder. When specified (see 6.2), a steel label holder finished in satin chromium plating and capable of holding a card or label approximately 2-3/4 inches wide by 1-1/2 inches high shall be provided. One label holder shall be provided for each set of two bunk bed panels.

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3.4.6 Fasteners. Machine screws for securing panels to bed shall be No. 12 or 1/4-inch diameter and of proper length to secure to the mating capped teenut as shown on figure 1. Teenut caps shall be nickle or chromium plated.

3.5 General construction. The design and dimensions of the bed headboard and footboard panels shall be as shown on figures 1 and 2 and as specified hereinafter. Unless otherwise noted, dimensions are minimum. Materials shall be in accordance with the details specified in 3.4 and hereinafter.

3.5.1 Panel construction. The panels shall be construction of particle board covered on each side with one piece decorative laminated plastic sheet. Adhesive for securing laminated plastic sheets to particle board shall be urea-resin type conforming to requirements of MMM-A-188. When label holders are specified, fasten a label holder with not less than three escutcheon pins to one of each set of two bunk bed panels at the location shown on figure 2. Finished laminated plastic sheets shall not be cracked, split, scratched, scuffed, peeled, dented, or contain ripples. The entire edge of each panel shall be covered with a tee-style plastic edge molding as specified in 3.4.3. The molding shall cover the panel edge including the laminated plastic sheets, or may cover the edge to within 1/64-inch of the edge. When necessary, the tongue shall be notched out at the corners of the panels for a snug, smooth fit around the panel radius. Each panel shall have but one edge molding joint, occurring approximately at the center of the bottom. Ends of the edge molding shall be square cut, meet flush, and be secured by brads at each side of the joint. The brads shall be driven from one side panel, set and filled to match the color of the laminated plastic sheet. The edging shall fit tightly and securely, without gaps or crevices. Finished plastic edge molding shall not be cracked, split, scratched, scuffed, or dented affecting serviceability or appearance.

3.5.2 Installation. When specified (see 6.2), the panel manufacturer shall install panels on headboards and footboards of the bunk beds at the military activity or activities specified in the invitation for bid or request for proposal. The footboard panels shall be installed outside of the cross boards, and the headboard panels shall be installed inside of the cross bars as shown on figure 1. All panels shall be centered between the corner posts.

3.5.3 Instruction sheets. Unless the manufacturer of the bed panels is required to install the panels on the bunk beds, two printed instruction sheets for insatllation of panels on the bunk beds shall be furnished in each carton of panels and in each bulk package of hardware (see 6.7). The sheet shall provide all necessary installation information and shall include a sketch of the completed unit and a cross-section view at the clamp similar to that shown on figure 1.

3.6 Workmanship. Workmanship shall be the highest quality throughout in accordance with the best standard practice used in the manufacturer of furniture of this type. There shall be no open joints affecting appearance. Exposed edges shall be staight and smooth, without protrusions. There shall be no sharp edges or corners exposed in the finished work which could be judged potentially harmful to wearing apparel or to the user.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one panel when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one panel. All all panels of the same size offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level (AQL) shall be 2.5 defects per hundred units (DHU) for major defects and 4.0 DHU for minor defects. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each panel selected shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table I. Each attribute within each classification of multiple defects shall constitute one defect.

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TABLE I. Classification of Defects.

Classification	Defects	Requirements Paragraph
Critical	None defined.	
Major:		
101	Panel not complete, components missing.	3.1
102	Overall dimensions or corner radius not as specified.	Figure 1
103	Particle board panel not as specified.	3.4.1
104	Laminated plastic sheets not as specified.	3.4.2
105	Plastic edge molding not as specified.	3.4.3
106	Clamps not as specified.	3.4.4
107	Label holder not provided, when specified.	3.4.5 and 3.5.1
108	Fasteners not as specified.	3.4.6
109	Metal finishes missing completely.	3.4.6
110	Plastic surfacing cracked, split, chipped, peeled, scratched, dented, or contains ripples.	3.5.1
111	Plastic edge molding surfaces cracked, split, scratched, scuffed, or dented affecting serviceability or appearance.	3.5.1
112	Molding edge and panel edge not within tolerance specified.	3.5.1
113	Poor fit of component parts.	3.6
114	Sharp edges or protrusions in plastic, wood, or metal surfaces that could cause injury or tear clothing.	3.6
Minor:		
201	Screw missing, cracked, or split or threads stripped.	3.4
202	Screw head broken off or evidence fastener is not of type or size specified.	3.4
203	Plastic surfacing not pattern, color or finish specified.	3.4.2
204	Plastic edge molding not pattern or color to match plastic surfacing as specified.	3.4.3

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TABLE I. Classification of Defects. Con't.

Classification	Defects	Requirements Paragraph
205	Joint in plastic edge molding in wrong location.	3.5.1
206	Panels not located on footboards and headboards as specified, when installation is required.	3.5.2
207	Instruction sheets not provided, when procured without installation.	3.5.3
208	Components not neatly attached, poor fit or mating part, or parts misaligned or uneven affecting appearance.	3.6

4.6 Packaging inspection. Preservation requirements shall be inspected in accordance with the requirements of MIL-P-116. Packing requirements shall be inspected in accordance with the requirements of MIL-STD-1186 and the applicable container specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level B or C as specified (see 6.2).

5.1.1 Level B. Each panel shall be wrapped with kraft paper conforming to UU-P-268 having a basis weight of 60 pounds. Prior to securing the wrap the instruction sheet, when furnished, shall be placed inside the wrap. The wrap shall be secured in place with tape or twine. Clamps and label holders, if not installed on the panel, shall be packaged in close fitting boxes and cushioned to prevent movement. The boxes shall be closed and secured with tape.

5.1.2 Level C. Panels, clamps, fasteners, label holders, and instruction sheets shall be packaged in accordance with the contractor's standard practice in a manner to prevent corrosion, deterioration, and damage during shipment.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The panels and accessories shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be cushioned, blocked and braced to prevent movement inside the container. The boxes shall be closed and strapped in accordance with the appendix to the box specification.

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5.2.2 Level B. The packaged panels and accessories shall be packed as specified for level A except the boxes shall conform to PPP-B-601, domestic type, PPP-B-621, class 1, PPP-B-636, class weather-resistant, or, PPP-B-640, class 2.

5.2.3 Level C. The panels and accessories shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulation.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The panels are for use as headboard or footboard panels with standard bunk beds conforming to MIL-B-15228. Two panels are required per bed. The panels clamp directly to the upper and lower cross tubes (7/8-inch O.D.) of bed head and foot.

6.2 Ordering data.

a. Title, number, and date of this specification.
b. Quantity required.
c. When a first article is not required for inspection and approval (see 3.2, 4.2.1, and 6.3).
d. Location of first article inspection (see 3.2 and 6.3).
e. Specify pattern, color, and finish of decorative laminated plastic sheets. Finish A (Satin) is generally used for solid-color plastic laminated, Finish B (Furniture) or Finish D (Textured) are appropriate for woodgrains (see 3.4.2).
f. Specify color of plastic edge molding when solid color is used with woodgrain pattern (see 3.4.3).
g. Specify color of PVC coating on clamps if color other than gray is required (see 3.4.4).
h. Specify label holders, when required (see 3.4.5).
i. Specify if panel manufacturer is required to install the panels on the bunk beds, and indicate location, quantity, and condition of beds at each location (see 3.5.2, 6.4, 6.5, and 6.6).
j. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample, a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one, unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Boring instructions. The distance between the top and bottom cross tubes of bunk bed headboards and footboards vary among beds procured from different manufacturers and between old design military standard bunk beds, and the new design. The distance between the cross tubes of the new design is

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3 inches more than the old design. Pre-drilling of panels for inserting the machine screws should not be accomplished by the panel manufacturer unless the dimension between the cross tubes of the footboard and headboard can be positively determined for all beds for which panels are being procured. When the panel manufacturer is required to bore panels, boring dimensions should be included in the invitation to bid or request for proposal. If more than one dimension is given, the number of panels for each boring requirement should be specified. When panels are not to be bored by the panel manufacturer, panels should be drilled and installed at the activity where installers have access to the bunk beds for accurate measurement.

6.5 Selection of panel installer. The selection of an installer of the bed panels at the least cost to the Government should be determined by weighing certain factors and selecting the one most applicable to the existing conditions. Installation may be accomplished by: (1) the panel manufacturer, (2) station personnel, or (3) local contractor. Selection of one of the above should be determined by considering the following, (a) quantity of panels to be installed, (b) are panels being procured in same contract or order with desks and chests, thereby making installation of panels more attractive to the manufacturer, (c) availability of station personnel, (d) availability and cost of contracting a local installer. An example of relatively small installation job would be an order of 200 pairs of panels. This would probably result in high installation cost if the panel manufacturer were required to install. In this situation, it would be more economical to install by station help or by local contractor.

6.6 Condition and location of bunk beds. When installation is required, specify condition and location of bunk beds so as bidder can submit bids accordingly. The following information should be provided

a. Are beds located in sleeping rooms or a central storage area?
b. Are beds assembled or knocked down (KD).
c. If KD, are bunk beds packaged or unpackaged. If packed, who is to unpack?
d. If beds are in sleeping rooms, will rooms be occupied during installation?
e. Will installation of panels be restricted to certain hours? days?

6.7 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.5.3	Instruction, Equipment Installation	UDI-E-21344	

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(Copies of Data Item Descriptions required by contractors in connection, with specific procurements functions should be obtained from the procuring activity or as directed by the contracting officer).

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensive changes.

Custodians:

Army - GL
Navy - YD

Review Activities:

Army - MD
Navy - MC

Preparing Activity:

Navy - YD

(Project 7105-0253)

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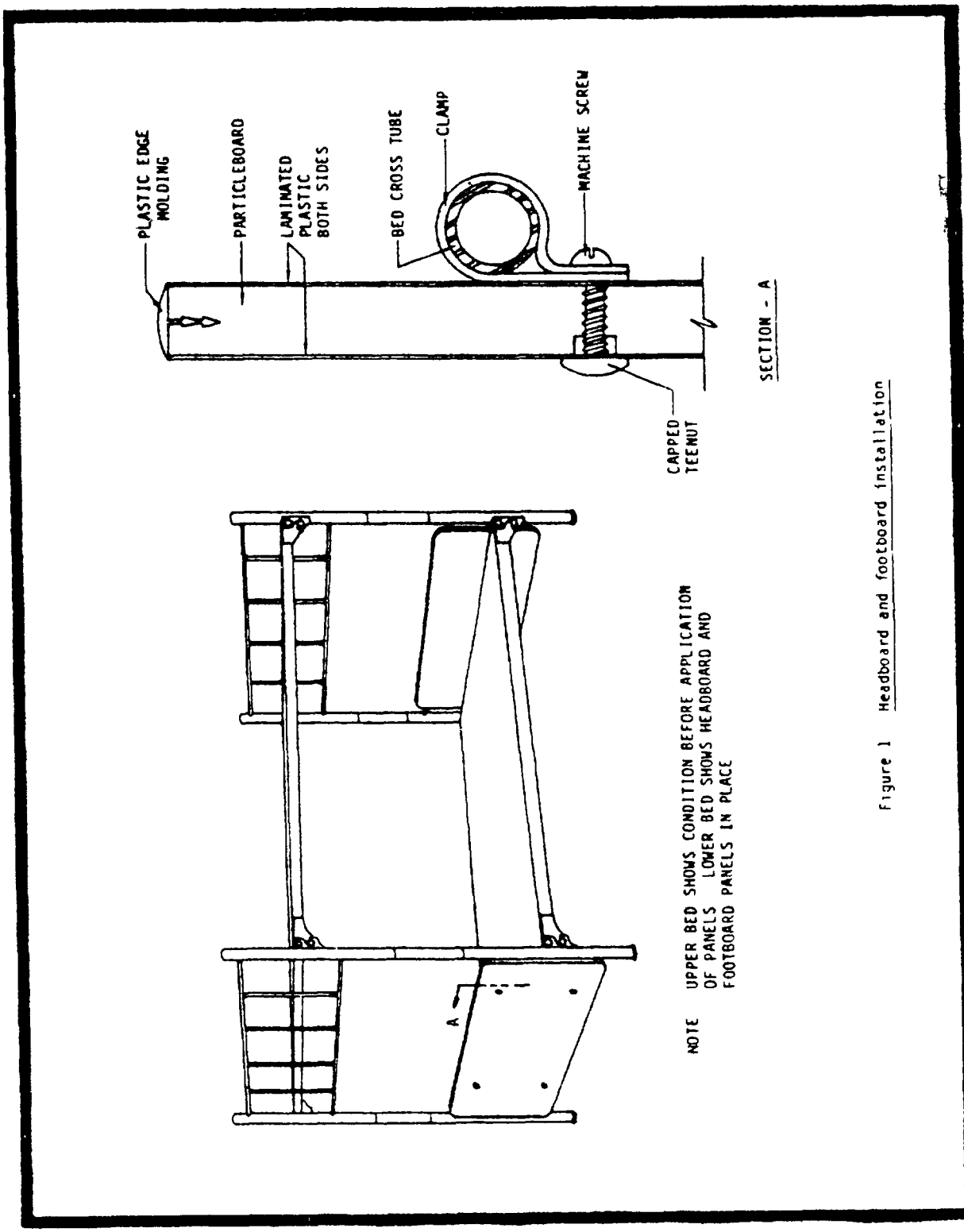


Figure 1 Headboard and footboard installation

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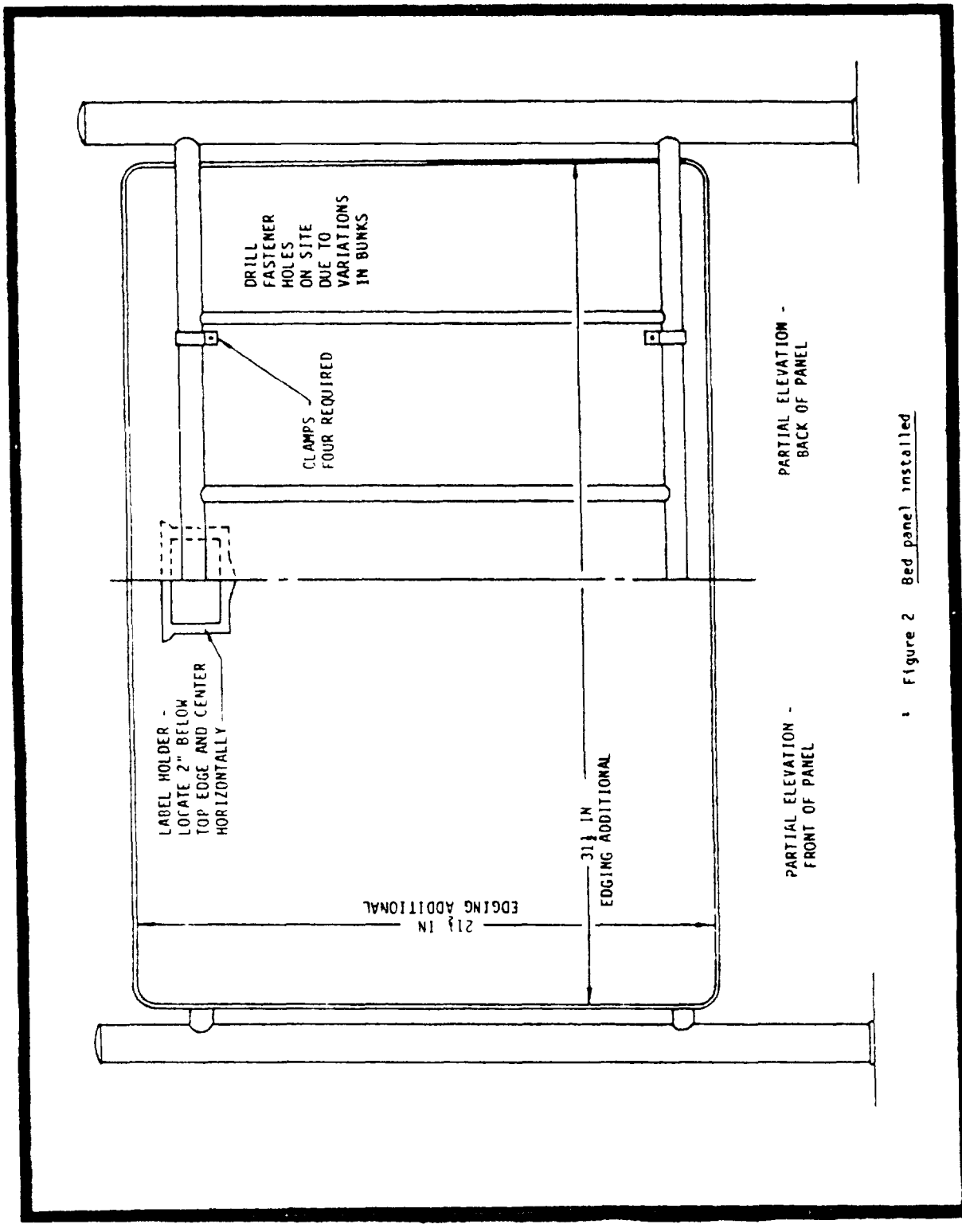


Figure 2 Bed panel installed

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-28636B	PANEL, FOR BUNK BED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street City State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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